

EXPERIMENT -05

AIM- To perform Simple Linear Regression with R

CODE-

```
import numpy as np import matplotlib.pyplot as plt from
sklearn.datasets import load_diabetes from
sklearn.linear_model import LinearRegression from
sklearn.model_selection import train_test_split from
sklearn.metrics import mean_squared_error,r2_score
```

```
#step 1 :Load the DataSet print("Loading
Dataset...") diabetes=load_diabetes()
```

```
#use only one features (BMI) to make it simple regression
x=diabetes.data[:,2].reshape(-1,1) y=diabetes.target
```

```
print(f"Dataset loaded with {len(x)} samples")
```

```
#split data into trianing and testing
```

```
x_train,x_test,y_train,y_test= train_test_split(
x,y ,test_size=0.2,random_state=42
)
```

```
#step-3: CREATE AND TRAIN MODEL
model=LinearRegression()
model.fit(x_train,y_train)
```

```
print("\n__ Model Training Complete__")
print(f"Slope (coefficient):{model.coef_[0]:.2f}")
print(f"Intercept:{model.intercept_:~.2f}")
```

#step-4 Make Predictions

```
y_pred=model.predict(x_test)
```

#step 5: Evaluate the model

```
mse=mean_squared_error(y_test,y_pred)
```

```
r2=r2_score(y_test,y_pred)
```

```
print("\n ---Model Performance---")
```

```
print(f"Mean Squared Error:{mse:.2f}")
```

```
print(f"R2 Squared :{r2:.2f}")
```

```
#visulisation the results plt.figure(figsize=(10,6))
```

#plot training data

```
plt.scatter(x_train,y_train,color='blue',alpha=0.5,label="training Data")
```

#plot test data

```
plt.scatter(x_test,y_test,color='Green',alpha=0.5,label="test Data")
```

#plot regression line

```
plt.plot(x_test,y_pred,color='red',linewidth=2,alpha=0.5,label="regressio  
n line")
```

```
plt.xlabel('BMI (Body Mass Index)') plt.ylabel('Disease
```

```
Prograssion') plt.title("Simple Linear Regression :BMI Vs Disease
```

```
Progression") plt.legend() plt.grid(True,alpha=0.3) plt.show
```

OUTPUT

```
Loading Dataset...
Dataset loaded with 442 samples

__ Model Training Complete__
Slope (coefficient):998.58
Intercept:152.00

---Model Performance---
Mean Squared Error:4061.83
R2 Squared :0.23

[3]: <function matplotlib.pyplot.show(close=None, block=None)>
```

